

Week - 4

Objectives:

- Understand and implement ES6 features such as arrow functions, callbacks, promises, and async/await.
- Perform asynchronous operations using the fetch() API to retrieve data from external sources.
- Integrate third-party APIs such as OpenWeatherMap into a web application.
- Parse and manipulate JSON data received from an API response.
- Visualize dynamic data using graphical representations with Chart.js.
- Design and develop a responsive web application using HTML, CSS, and JavaScript.

AIM: Implement an application for reading the weather information from openweathermap.org and display the information in the form of a graph on the web page.

Index.html:

```
<!DOCTYPE html>

<html>

<head>

    <title>Weather Graph App</title>

    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

    <link
href="https://cdnjs.cloudflare.com/ajax/libs/bootstrap/5.3.0/css/bootstrap.min.css"
rel="stylesheet">

    <script
src="https://cdnjs.cloudflare.com/ajax/libs/bootstrap/5.3.0/js/bootstrap.bundle.min.js"></script>

</head>

<body style="background: linear-gradient(to bottom, #aaf2ff, #ffffff); min-height: 100vh;">

    <div class="container mt-5">

        <h2 class="text-center mb-4">WEATHER GRAPH WEBPAGE</h2>

        <div class="input-group mb-3">

            <input type="text" id="city" class="form-control" placeholder="Enter the City Name"

        />

        <button class="btn btn-primary" onclick="getWeatherInfo()">Click Here for Weather
Info</button>
```

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Experiment No: _____ Date: _____

```
</div>

<div class="card">
  <div class="card-body">
    <canvas id="weatherchart" width="400" height="300"></canvas>
    <div id="weatherInfo" class="mt-3"></div>
  </div>
</div>

</div>

<script>
  const api_key="86583a4fa46e0c89d17d27cd117ff9d7";

  const getWeatherInfo=async()=>{
    const city=document.getElementById("city").value;
    try{
      const res=await
fetch(`https://api.openweathermap.org/data/2.5/forecast?q=${city}&units=metric&appid=${
api_key}`);
      const data=await res.json();
      const lab=data.list.slice(0,12).map(item=>new
Date(item.dt_txt).getHours()+":00");
      const temp=data.list.slice(0,12).map(item=>item.main.temp);
      disp(lab,temp);
      const dt=data.list.slice(0,12);
      const
avgHu=(dt.reduce((sum,item)=>sum+item.main.humidity,0)/dt.length).toFixed(2);
      const
avgP=(dt.reduce((sum,item)=>sum+item.main.pressure,0)/dt.length).toFixed(2);
      const
avgw=(dt.reduce((sum,item)=>sum+item.wind.speed,0)/dt.length).toFixed(2);
      document.getElementById("weatherInfo").innerHTML = `
<div class="mt-3 p-3 bg-light rounded">
  <h5>36 Hour Average</h5>
```

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```
<p><strong>Avg Humidity:</strong> ${avgHu} %</p>
<p><strong>Avg Pressure:</strong> ${avgP} hPa</p>
<p><strong>Avg Wind Speed:</strong> ${avgw} m/s</p>
</div>

};

}catch(err){
    console.error("Error fetching weather:",err);
}
};

const disp=(lab,data)=>{
    const ctx=document.getElementById('weatherchart').getContext('2d');
    new Chart(ctx,{
        type:'line',
        data:{
            labels:lab,
            datasets:[{
                label:'Temperature(°C)',
                data:data,
                borderWidth:2
            }]
        }
    });
};

</script>
</body>
</html>
```

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Experiment No: _____ Date: _____

OUTPUT:

